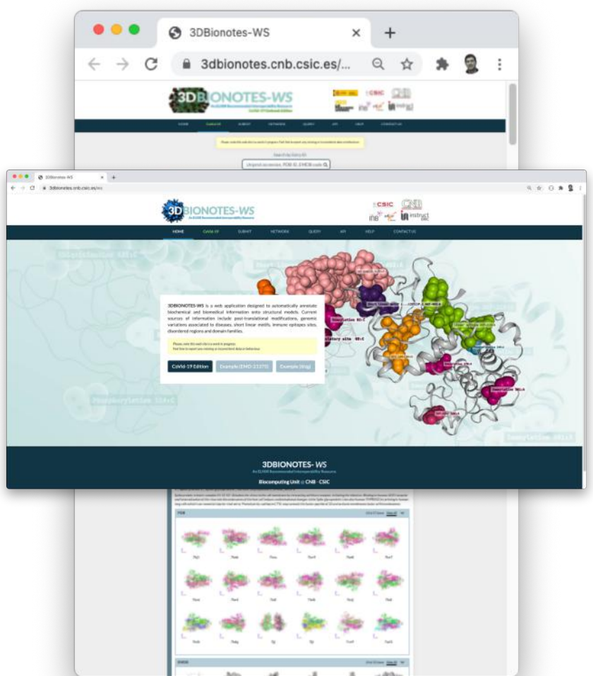
Updates On 3DBionotes-WS and Future Plans

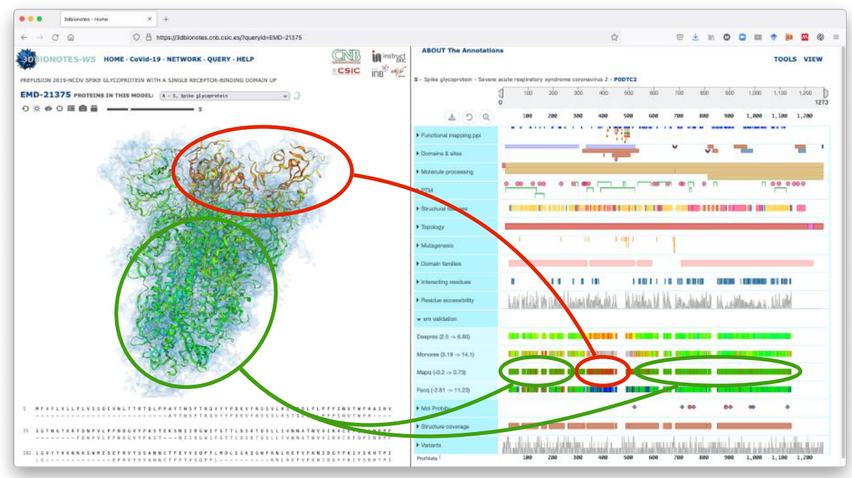
Jose Ramon Macias, Erney Ramírez-Aportela, Jose Luis Vilas, Marta Martinez Gonzalez, Carlos Oscar Sanchez Sorzano, Jose Maria Carazo

Spanish National Bioinformatics Institute (INB ELIXIR-ES). Biocomputing Unit, National Centre of Biotechnology (CNB-CSIC). Instruct Image Processing Centre

The 3DBionotes-WS web platform (Macias et al., 2021) has been one of the recommended interoperability resources by ELIXIR for several years. A set of Web Services providing multiple annotations oriented to structural biology are integrated in a website interface including a 3D viewer for macromolecular models and Cryo-EM derived maps. The 3D viewer is fully interactive with a corresponding display for genomic, proteomic and structural feature annotations.

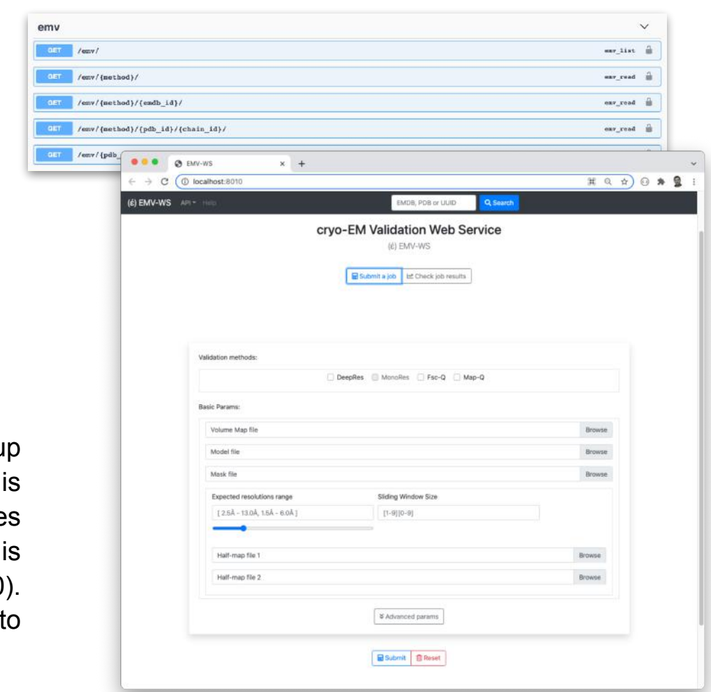


Achievements in Cryo- Electron Microscopy (cryo-EM) and other techniques of structural biology, instrumentation, sample preparation and image processing methods allowed for an increase in the resolution obtained, as well as in the number of resolved structures deposited at public databases like MDB (Patwardhan, 2017). That was remarkably obvious with the efforts to generate key viral structures to enable the early development of vaccines and potential antiviral drugs during the COVID-19 outbreak. However, some concerns have been raised to ensure that data quality is also maintained both globally for the entire structure and locally in critical areas of the model.



To evaluate the quality of cryo-EM maps and their fitted atomic models an additional group of annotations has recently been incorporated into the 3DBionotes. Map quality is evaluated by local resolution using DeepRes (Ramírez-Aportela et al., 2019), BlocRes (Cardone et al., 2003) and MonoRes (Vilas et al., 2018); while the map-to-model fit is examined with FSC-Q (Ramírez-Aportela et al., 2020) and Q-score (Pintilie et al ., 2020). The values obtained are projected on the atoms of the atomic model, using color codes to represent the different thresholds.

These new set of annotations is also served for further integration by third party tools through a convenient Application Programming Interface (API). Currently this validation and quality measurement data is only available for those map and models already deposited in public repositories (EMDB, PDB). As a future enhancement, we are working to set up a new web service that allows users to evaluate their maps and models through a simple web interface before submitting them to those repositories.



Cardone, G., Heymann, J. B. & Steven, A. C. One number does not fit all: mapping local variations in resolution in cryo-EM reconstructions. J. Struct. Biol. 184, 226–236 (2013).
Macias J.R., Sanchez-Garcia R., Conesa P., Ramirez-Aportela E., Martinez Gonzalez M., Wert-Carvajal C., Parra-Perez A.M., Segura Mora J., Horrell S., Thorn A., Sorzano C.O.S. Carazo J.M.. 3DBionotes COVID-19 edition, Bioinformatics, 2021; btab397, https://doi.org/10.1093/bioinformatics/btab397.
Patwardhan, A. (2017). Trends in the electron microscopy data bank (emdb). Acta Crystallographica Section D: Structural Biology, 73(6).
Pintilie, G., Zhang, K., Su, Z., Li, S., Schmid, M. F., & Chiu, W. (2020). Measurement of atom resolvability in cryo-EM maps with Q-scores. Nature Methods, 17(3), 328–334.
Ramírez-Aportela, E., Mota, J., Conesa, P., Carazo, J. M., & Sorzano, C. O. S. (2019). DeepRes: a new deep-learning- and aspect-based local resolution method for electron-microscopy maps. IUCrJ, 6(Pt 6), 1054–1063
Ramírez-Aportela, E., Vilas, J. L., Glukhova, A., Melero, R., Conesa, P., Martínez, M., Maluenda, D., Mota, J., Jiménez, A., Vargas, J., Marabini, R., Sexton, P. M., Carazo, J. M., & Sorzano, C. O. S. (2020). Automatic local resolution-based sharpening of cryo-EM maps. Bioinformatics , 36(3), 765–772.
Vilas J.L., Gomez-Blanco J., Conesa P, Melero R., de la Rosa-Trevin J.M., Oton J., Cuenca J., Marabini R., Carazo J.M., Vargas J., Sorzano C.O.S. MonoRes: automatic and accurate estimation of local resolution for electron microscopy maps. Structure 26, 337–344 e334 (2018).

https://3dbionotes.cnb.csic.es/ws/